

Chapter 1 /CO1

(4m)

Describe the “platform-independent” feature in Java.

Describe primitive and non- primitive data types of java with appropriate code snippets

Explain how to take user input in Java using the Scanner class with a suitable example program.

Explain the platform-independent feature of Java — how Java achieves “write once, run anywhere” — with a suitable example.

Explain Bitwise Operators with example.

(5m)

Classify data types of java with appropriate code snippets

Explain the use of the conditional (ternary) operator in Java and illustrate it with an example

State the significance of Java Virtual Machine (JVM) in the Java programming environment.

Define type casting and explain two types of type casting with suitable example. Explain the bitwise operator with an example.

Explain the role and importance of the Java Virtual Machine (JVM) in the Java programming environment.

Chapter 2 /CO2

(4m)

Describe the working of the “this” keyword with an example.

Describe the working of new operator in java with appropriate example

Explain Method Overloading in Java with a suitable example program and output.

Explain the meaning of the final keyword with a simple program.

Explain the working and purpose of the static in Java with a suitable example and output.

(5m)

Describe the process of creation of objects in java with suitable example.

Explain static data and static member method with example program

Define constructors and explain default and parameterized constructor with example

Describe the method over riding in java with suitable example.

Write a Java program to demonstrate the use of static data member and static method. Explain the output of your program.

(8m)

Write a Java Program to create a class known as Shape with data members as radius and height. Use getter and setter methods. Create two subclasses named as Cone (with method named as volumOfCone) and Cylinder (with method named as VolumeOfCylinder). Write a Java Program to Calculate Volume of Cone and volume of cylinder based on formula (formula: Volume of cone: $\frac{1}{3}\pi r^2 h$, Volume of Cylinder: $\pi r^2 h$).

Explain the following methods in the String class with an example. i) charAt() ii) replace(char oldchar, char newchar) iii) trim() iv) indexOf(String str)

Explain the following methods in the String class with an example. i) length() ii) equals(String anotherString) iii) startWith(String prefix) iv) endsWith(String prefix)

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Chapter 3

(4m)

Compare method overloading and method overriding. (min 4 points)

Define Interface. Describe implementation of interface with appropriate example

Explain the final keyword in Java, its uses with variables with a suitable example.

Describe Enum and explain any five inbuilt packages in Java.

Explain Abstract class with example.

Compare abstract class, interface and class

Differentiate between Method Overloading and Method Overriding

(5m)

Describe Polymorphism and explain its types with suitable example.

Describe Inheritance

Explain Run time Polymorphism with suitable code snippet

Explain how to create a package and import it with suitable example.

Implement a class named Student that demonstrates the concept of constructor overloading . The class should store basic student information and provide multiple constructors.

Explain the purpose of packages in Java. Discuss any five commonly used inbuilt packages with their functionalities.

Explain how to create a package and import it with suitable example. implementation of interface with appropriate example

Write a Java program to create a class called Shape with a method called getArea(). Create a subclass called Rectangle that overrides the getArea() method to calculate the area of a rectangle.

Design a Java program demonstrating method overriding.

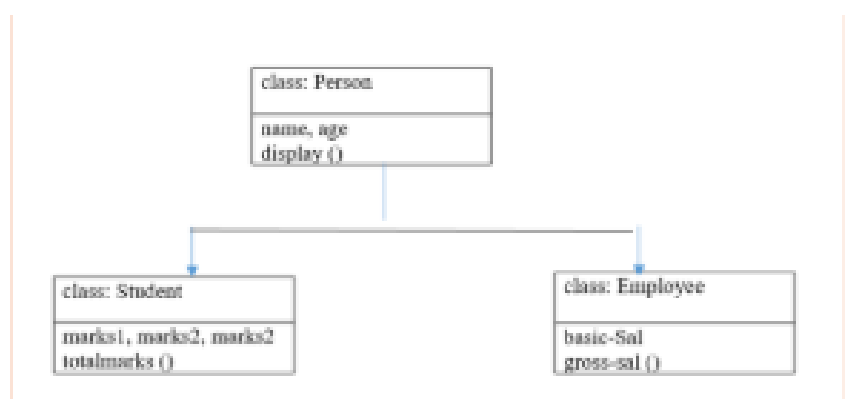
(8m)

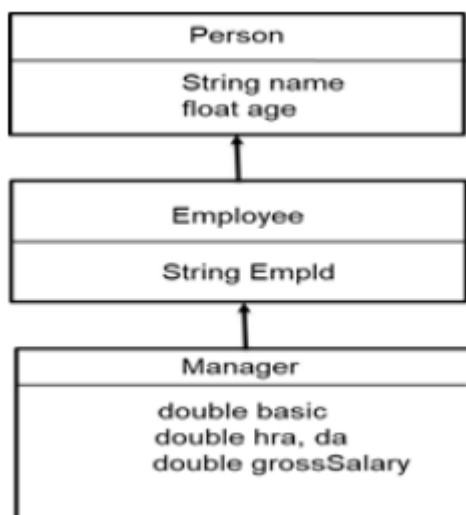
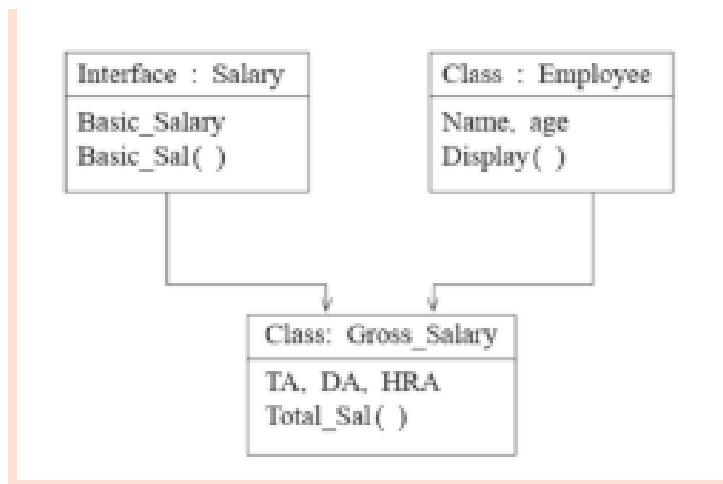
Consider a class named "Student" having id and name as instance members. A class "Test" inherits Student and having PT1 and PT2 as data members. There is one interface named "Sports" which contains a variable points assigned with value 5. (additional points of sports should be added in total marks of student who participated and won in sports activities in the semester)

Write a Java program to calculate the total marks of a student.

Write a Java program to create a class called Employee with methods called work() and getSalary(). Create a subclass called HR Manager that overrides the work() method and adds a new method called addEmployee().

Implement the following Inheritance Implement the following Inheritance





Write a Java program to implement the Inheritance as shown in the figure. Create an instance of Manager class. Read basic salary of a Manager and calculate its gross salary. Display the information of a Manager. (Note: $\text{hra} = 10\%$ of basic, $\text{da} = 8\%$ of basic $\text{grossSalary} = \text{basic} + \text{hra} + \text{da}$)

Chapter 4 CO4

(4m)

Explain Checked and Unchecked exceptions.

Draw and describe the class hierarchy of Exception. Write a Java program to find out sqrt of a number, generate an exception for a negative number

Explain the class hierarchy of exceptions in Java with a neat diagram and suitable examples

Write a Java program to find out sqrt of a number, generate an exception if any.

Describe the purpose and usage of the following keywords in Java with suitable code snippets: a) try b) throw c) throws d) catch e) finally Explain checked and Unchecked exceptions in Java

(6m)

WAP in Java to read Employee experience (years) (lower limit = 0 upper limit = 40) and generate an exception if it is 41 or above

Explain the process of creating a user-defined exception with an example program

Develop a Java program to input an employee's experience. The experience should be between 0 and 40 years. If a value greater than 40 is entered, the program must generate and handle an exception.

WAP in Java to read Employee salary (lower limit = 10000 upper limit = 100000) and generate an exception if salary is out of range

Write a Java program to read the employee's experience in years. The program should have a lower limit of 0 and an upper limit of 40 for the experience. If the input value is 41 or above, the program should generate a custom exception.

Chapter 5 (CO5)

(4m/5m)

Explain the life cycle of a thread in Java with a neat diagram. What are the different thread states?

Explain inter-thread communication in Java with its methods and a suitable example program.

Explain the different ways to implement threads in Java with suitable code snippet for each method.

Discuss the various phases a thread undergoes from creation to termination in Java, along with a suitable example.

Explain the methods Wait(), Notify() ,notifyall()

Describe Thread Synchorization.

Explain the Use wait() and notify() methods for communication between threads.

(6m)

Design a Java program using multithreading where one thread (Producer) generates a series of numbers and another thread (Consumer) consumes them. The producer should not produce if the buffer is full, and the consumer should not consume if the buffer is empty. Use wait() and notify() methods for inter-thread communication.

Write a Java program where the main thread waits for two child threads to complete execution using the join() method.

Describe the role of synchronization in multithreaded Java programs. Explain how synchronized methods and synchronized blocks work to prevent thread interference

Chapter 6 (CO6)

(4m)

Compare AWT and Swing in Java based on the following parameters:

- a) Platform Dependency b) Look and Feel c)Components and d) Performance.

Explain the use and syntax of the ActionListener interface in Java with suitable code snippet.

Explain the use and syntax of the MouseListener interface in Java with suitable code snippet.

Describe the KeyListener interface in Java along with its methods and a suitable example program.

(8m)

Write a Java program using AWT to create a login form. The program should validate the entered username and password and display a message indicating whether the login is successful or failed

Write a Java program to design a simple Login Form using Swing components. The program should accept a username and password from the user and display an appropriate message indicating whether the login is successful or unsuccessful.

Write a Java program to implement a simple arithmetic calculator that performs basic operations like addition, subtraction, multiplication, and division.

WAP to design a Login form in Java and display if the login is successful or not.

Write a Java program to design a simple arithmetic calculator using Swing components. The program should allow the user to enter two numbers and perform basic arithmetic operations such as addition, subtraction, multiplication, and division.

Develop a AWT Frame to have 2 textboxes for entering units and displaying the calculated electricity bill. The bill should be calculated as per below details:

Consumption Units	Rate of Charge
0 - 180	Rs. 40 plus 60 Paisa per unit
181-240	Rs. 80 plus 80 paisa per unit excess of 180
241-300	Rs. 120 plus 90 paisa per unit excess of 240
Above 300	Rs. 160 plus Rs. 1 per unit excess of 300

Develop a AWT Frame having labels and text boxes to enter the name, grade and basic salary of an employee. Provide a label and text box to display the gross salary. Each employee IS getting Travelling Allowance and Entertainment Allowance as specified in the below table. The FIRA is given as 35% of basic pay to all the employees. Add 2 buttons i.e. "CalGrossSal" and Reset to calculate the gross salary and to reset all the components respectively.

Grade	TA(Rs.)	EA(Rs.)
A	2500	2000
B	2000	1500
C	1500	1000
D	900	800

(6m)

Difference Between AWT and Swing. Give the different Swing controls. Also, write a simple program to demonstrate Swing controls.

Give the different ways to create a thread in Java? Explain with examples Develop a Java program where a user types text into a TextField and the program displays the last key pressed in a label.

Write a Java program using Swing components. Also, explain how Swing is different from AWT and list the various Swing controls available.

Explain the various approaches to thread creation in Java. Provide example programs for each approach.